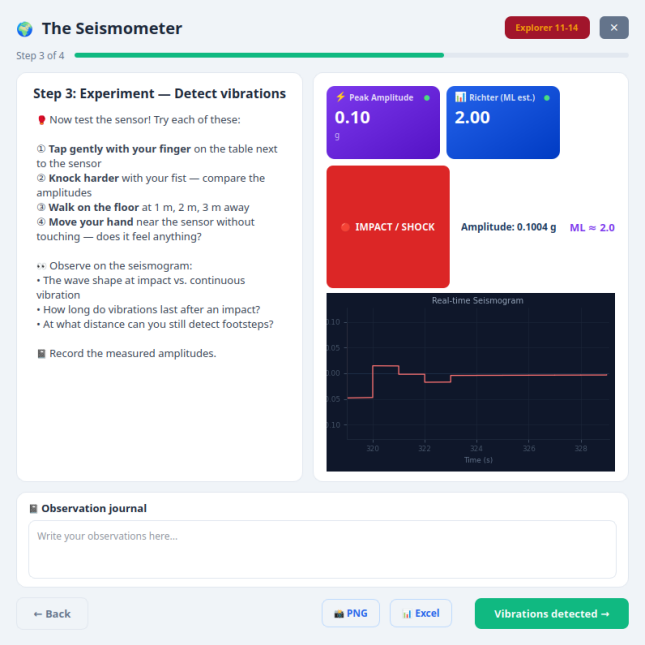
The Seismometer

A box that feels footsteps in the corridor — and the difference between magnitude and intensity, measured

Level: Explorer — ages 11–19 (Romanian school years V–XII) • **Time in class:** 20 min, or one 50-minute lesson • **Sensors:** MPU-6500 (acceleration)



The activity screen. The values in the picture are simulated; your own will differ.

Before you start

Who it is for

Explorer — ages 11–19 (Romanian school years V–XII). As a demonstration it works at any age. The most direct entry is geography: **year V** (earthquakes) and **year IX** (seismicity and the Earth's internal structure), because the activity makes the difference between magnitude and intensity *measurable*. In physics, inertia (lower secondary) and acceleration (year IX).

Assumed knowledge

The idea of acceleration and of vibration. Reading a graph that oscillates about zero. Logarithms are **not** needed for the basic version; they appear only if you discuss how a logarithmic scale is built.

Time in class

Short: 20 min — the wizard's seven steps. Full: one 50-minute lesson, with the series of vibration sources and the discussion of magnitude against intensity.

Teacher preparation

10 min — choose a **stable** table that does not wobble, and test it: a flimsy table puts every footstep in the room onto the graph and ruins any comparison. Mark the box's position and the striking distances with tape. And run a box for two minutes to see how quiet the room is.

Sensor settling time

Ten seconds of complete stillness before every measurement — **and this is the hardest thing in the activity**, because it means nobody touches the table, nobody leans on it and nobody walks past it. The app's filter needs a few seconds to learn where zero is.

Sensors and boxes	MPU-6500, the accelerometer. One box is enough for the whole class with the screen projected — and that is the recommended version, because it keeps thirty pairs of hands away from the table.

Three routes

Short	20 min: the wizard's seven steps — how a seismograph works, a prediction, calibration on silence, a second prediction, detecting vibrations, *How sure are we?*, the conclusion.
Full	The 50-minute lesson: the short route plus **the series of sources** — a light tap, a hard knock, a footstep at one metre, a footstep at five, a door closed — and the discussion of why the same event gives different numbers at different distances.
Project	Extension: the same source measured from three distances, so that the fall-off with distance becomes visible — the direct bridge to the *Seismology* app, with several boxes.

1. Where this fits in the curriculum

- **Geography** — in **year V**, earthquakes as a phenomenon of the Earth's crust, and in **year IX**, seismicity and the Earth's internal structure. This is the most direct entry in the whole collection, because the activity does not merely illustrate the difference between magnitude and intensity but makes it *measurable*: the same knock gives different numbers at different distances. How far the syllabus goes with the seismic scales varies from year to year — check the current syllabus.
- **Physics** — inertia in lower secondary, and acceleration in **year IX**. A seismograph is inertia turned into an instrument: the mass wants to stay put while the frame moves with the ground.
- **Mathematics** — logarithms are taught in **year X**; from there on you can discuss why seismic scales are logarithmic. For years V–VIII the rule in words is enough: one unit up means ten times the ground motion.
- **Emergency preparedness** — the natural bridge to the *Earthquake!* activity, which is about safe behaviour. **In Romania this is not a theoretical topic**: the Vrancea earthquakes are a reality every class lives with.
- **Cross-curricular** — the idea of a *noise floor*: an instrument never measures "nothing", it always measures something small, and knowing how small that something is is the condition for being able to say you have detected something large.

The core skill is **telling what happened from how much it was felt**. An earthquake has a single magnitude but as many intensities as there are places. The box measures the second, not the first — and that is precisely what makes it a good teaching instrument.

2. Time plan

Route	Time	What happens
Short	**20 min**	How a seismograph works (3), the first written prediction (2), calibration on silence — that is, measuring the noise floor (4), the second prediction (1), detecting vibrations (6), *How sure are we?* (2), the conclusion (2). These are exactly the wizard's seven steps.
Full	**50 min**	The short route unhurried (20), the series of vibration sources from a light tap to a closing door (15), the distance series for one source (10), the discussion of magnitude against intensity (5).
Extension	+20 min	The same source measured from three places in the room, with the box moved — direct preparation for the *Seismology* app, where the measurement is made simultaneously with several boxes.

Silence is the hardest equipment to obtain

This activity needs something a classroom almost never has: **ten seconds in which nobody touches the table**. An elbow resting on it, a foot knocking a table leg, a classmate walking past — all appear on the graph, larger than the event you are trying to measure.
Three things solve it, and all three must be done. **The table is declared out of bounds**: nobody touches it except the person striking. **The measurement is announced aloud** — "quiet, measuring" — and the ten seconds are counted aloud, so that everybody knows how much longer. And **the class stays seated**, because thirty people standing on a wooden floor are a permanent earthquake.
Paradoxically, this is the most valuable part of the activity: pupils discover for themselves why real seismographs are installed in basements, on bedrock, far from roads.

3. Materials

Item	Per group	Notes
MagicBox with MPU-6500	1	The accelerometer. **One box with the screen projected** is the recommended version, not a fallback.
Laptop with the app	1	USB or WiFi. The cable must not hang off the table — it pulls the box and creates vibrations of its own.
A stable table	1	**Tested beforehand:** a wobbling table makes any comparison impossible. If you have none, a concrete floor beats a bad table.
Marking tape	1 roll	For the box's position and the striking distances — 0.5 m, 1 m, 2 m, 5 m.
A ruler or tape measure	1	The distances are measured, not judged.
Vibration sources	—	A fist, a pencil, a book dropped from a fixed height, a door closed normally. **Nothing that strikes the box itself.**
A piece of foam or sponge	1	Optional but excellent: the box on foam shows how much a soft foundation attenuates — exactly what engineers do with buildings.

Preparation. Choose and *test* the table: put the box down, start the activity and walk normally around the room. If the graph jumps at every step, the table is too light or the floor too springy — move before the lesson. Mark the box's position and the distances with tape. And leave a box for two minutes on the table, in silence, to see **the room's noise floor**: it is the figure everything else will be compared against.

4. Scientific background for the teacher

How inertia turns a box into a seismograph

A classical seismograph is a mass suspended from a frame fixed in the ground. When the ground moves, the frame moves with it, but **the mass wants to stay where it is** — inertia. The *relative* motion between frame and mass is recorded, and it is an image of the ground's motion.

The accelerometer in the box does the same thing, microscopically and electronically: a tiny proof mass held in silicon springs shifts relative to the frame when the box accelerates, and the shift is read capacitively. **It is, literally, a seismograph on a chip.**

An accelerometer at rest does not read zero, though: it reads **gravitational acceleration**, about 1 g on the vertical axis. The app therefore has to subtract gravity before it can draw anything useful — see the technical note — and what remains, oscillating about zero, is the *seismogram*.

4.1 Magnitude and intensity — the distinction the activity measures

This is the central idea of the activity, and the one the geography lessons on earthquakes rest on.

Magnitude describes **the source**: how much energy the rupture at depth released. Today the **moment magnitude**, written M_w, is used; the old "Richter scale" (local magnitude, M_L) is its

ancestor and is no longer the scale large earthquakes are reported on. An earthquake has *one* magnitude, however many people feel it and wherever they are. To compute it you need the recorded amplitude *and* the distance to the epicentre, because every recording has to be corrected for how much the wave died away on the journey.

Intensity describes **what was felt in one particular place**: how hard it shook, what broke, what people felt. Romania and the rest of Europe use the **European Macroseismic Scale, EMS-98**, with twelve degrees written in Roman numerals; the Mercalli scale is its older relative, still used in the Americas. A single earthquake has *as many* intensities as there are places, and they fall off with distance.

The box measures intensity, not magnitude. It sees the shaking at one point — its own — and has no way of knowing how far away or how deep the event was. This is not a limitation to be fixed but precisely what makes the activity useful in geography: the same knock, measured from one metre and from five, gives two different numbers *although the event was the same*. If pupils understand why, they have understood the distinction.

4.2 Why the wave dies away with distance

Two mechanisms, and both show up in the distance series.

Geometrical spreading: energy set off from a point spreads over an ever-larger surface, so the amplitude falls even if nothing is lost. For waves travelling through the volume, the fall is roughly as $1/r$.

Absorption: the material turns part of the wave's energy into heat. A carpet, a hollow wooden floor or a layer of foam attenuates far more strongly than concrete — and this is exactly why **the foam test is worth doing**: modern buildings in seismic zones are sometimes set on isolators precisely to exploit the effect.

At the scale of a classroom, a knock measured from five metres typically gives **five to twenty times less** than the same knock from half a metre. With a noise floor of about 0.01 g, that means a source must give at least 0.1 g close up to still be seen from five metres; if the far reading is lost in the noise, the ratio you compute is only a lower bound — and "from what distance can it no longer be seen" is itself a good result.

4.3 Logarithmic scales: why one unit means ten times

Seismic amplitudes span many orders of magnitude, from nanometres to metres. No linear scale can hold them, which is why every magnitude scale is **logarithmic**: one unit up means ground motion **ten times** larger (and, in energy, about thirty-two times).

It is worth saying in class with a Romanian example: the Vrancea earthquake of 4 March 1977 had a magnitude of M_w 7.4, and one of 6.4 — a single unit smaller — means motion ten times weaker. **On this scale, the difference between "large" and "catastrophic" is one unit.**

5. Lesson flow

The in-app wizard has **seven steps**: how a seismograph works, a prediction, calibration on silence, a second prediction, detecting vibrations, **How sure are we?** and the conclusion. **Before each measuring step the wizard asks for a written prediction** (button **My prediction is written**), which it quotes back in the conclusion; the pupil journal, predictions included, is saved to disk.

1. **How a seismograph works (3 min).** The app's first step. Explain inertia with a gesture: hold a book on your palm and move your hand sharply — the book lags behind. That is the proof mass. The button is **Got it!**

2. **The first prediction (2 min).** The app asks: *with nobody touching the table, will the line be perfectly flat? How big do you think the sensor's own noise is?* The answer goes into the prediction box.

Technical note — the filter, the fixed scale, and the shaking index

The sensor. MPU-6500, a three-axis accelerometer. **The motion sensor is mounted upside down inside the box, and the firmware already corrects for it**: a box lying flat reports $AZ \approx +1$ g. If yours reads -1 g flat, either it is running old firmware and needs reflashing, or its sensor is fitted the other way up and needs **Motion sensor upside down** ticked under **Settings → Box names** — it is not something to "correct" inside the activity. The app reads **every sample** (50 a second), so no short knock slips between two readings.

The filter in the chip. From firmware 2.2.5, the chip smooths the accelerations itself with a low-pass filter of about 20 Hz: vibrations below about 25 Hz — footsteps, doors, knocks on the table — are reproduced faithfully, and the fast hum of motors and fans no longer shows up as false trembling. The peak of a very short knock comes out somewhat smaller than with the old firmware, so **when you compare boxes, they must all run the same firmware**.

What is on the screen. Above the chart, the shaking state — **CALM** (below 0.012 g), **LIGHT VIBRATION** (up to 0.03 g), **MODERATE VIBRATION** (up to 0.1 g), **STRONG SHAKE / IMPACT** (above 0.1 g) — and an **events detected** counter, which counts peaks above 0.03 g separated by at least half a second of calm. The chart is called **Shaking (gravity removed)**; the green band around zero is the calm zone. Below it, two cards: **Peak amplitude**, in `g`, and **Shaking index**, in `log Gal`. The **Save the peak** button records the peak shown on the card; in the calibration and detection steps it must be pressed at least once before the wizard lets you move on, and the saved values appear in the conclusion.

The filter that removes gravity. An accelerometer at rest reads 1 g, not zero. The app subtracts a *slow average* of the signal **separately on each axis** — a high-pass filter — and draws only the departures from it, combined from all three axes. That way gravity disappears however the box is lying, and a sideways shake counts as much as a vertical one. Two practical consequences follow: the line sits near zero although gravity is there, and **the filter needs a few seconds of quiet to learn where zero is**. If you start measuring while somebody is shaking the table, the zero line is learned wrongly.

The sensor's own noise. Even on a perfectly quiet table the line trembles by about ± 0.01 g: this is the chip's electronic noise, measured on our boxes. Below about 0.012 g the app shows the **calm** state; only above that threshold can one speak of a vibration.

The window and the scale are fixed. The graph shows the last **ten seconds** and has a *fixed* vertical scale of ± 0.25 g. The fixed scale is a good choice — it makes two measurements comparable — but it means **a very hard knock runs off the graph**. If the peak reaches the edge, strike more gently: the card's value stays usable, but the shape of the wave is lost.

The peak is held for a few seconds before it starts to decay, so that you have time to read it. It is not an instantaneous value.

The shaking index, in the class's words. *Gal* is a unit of acceleration used by seismologists: 1 Gal = 1 cm/s^2, and 1 g $\approx$ 981 Gal. The app takes the peak in Gal and tells you *how many zeros it has*: 10 Gal gives 1, 100 Gal gives 2. For pupils aged 11–14 the rule is enough: **each unit up means shaking ten times stronger**. The proper word — base-10 logarithm — is taught from year X. Below the calm threshold the app shows 0, because there the number would only describe the sensor's noise; that is why the index jumps straight from 0 to about 1.1. A few reference points:

Peak	in Gal	Index
below 0.012 g (calm)	below 12	0
0.012 g	≈ 12	1.1
0.03 g	≈ 30	1.5
0.1 g	≈ 100	2.0
0.25 g (edge of the chart)	≈ 245	2.4

That formula has no distance correction, and the distance correction is exactly what turns a measured amplitude into a magnitude. Therefore:

- the number **is not a magnitude** and is not comparable with the magnitudes in the news;
- what it measures is *the shaking at the sensor*, a quantity akin to **intensity**.

Use the index as a relative scale, to compare knocks with each other; the difference between magnitude and intensity remains the question of the lesson.
Exporting the data. At the measurement steps, **PNG** (saves all the step's charts) and **Excel** appear. The Excel file holds the step's **raw, unfiltered readings** — the three axes, gravity still in them — and an `elapsed_s` column with the seconds since the step began; the filtered seismogram is not exported. **There is no CSV button inside the activity**; the complete files are taken from the **Data & Export** page. **The IMU file is separate** from the slow-sensor one, because it is sampled at a different rate — look for the `_imu` suffix.

3. **Calibration — that is, measuring the silence (4 min).** The box on the table, nobody touching it, ten announced seconds of quiet. Press **Save the peak** and note the peak value during the silence: this is *the room's noise floor* — about 0.01 g on a good table — and everything that follows is compared with it. The button to move on is **Baseline stable**. **It is the step classes skip and the one that makes the rest mean anything.**

4. **The second prediction (1 min).** *Will the box feel a footstep taken five metres away? What about a door closed across the room? And if I knock twice as hard, will the number be twice as large?* The third question prepares the logarithmic scale.

5. **Detecting vibrations (6 min).** In turn, with silence between them: a light tap with a finger at one metre, a fist at one metre, a footstep at one metre, a footstep at five metres, a door closed normally, each one three times. After each, press **Save the peak**, write in the journal what it was and compare the peak with the noise floor. The button to move on is **Vibrations detected**.

6. **How sure are we? (2 min).** The app shows the saved noise floor, the saved peaks and their spread, and reminds the class that a peak counts only if it is at least two or three times the floor. The button is **Continue**.

7. **The conclusion (2 min).** The final step quotes the prediction, shows the noise floor, the largest peak with its index and how many times the peak exceeds the floor, then links the activity to real seismology. Close with the question that matters: *the same knock gave* 0.08 *g from one metre and* 0.02 *g from five. Which number is "true"?* Both — because both measure how much was felt there, not how hard the knock was.

The investigation, in one table

The question. How does the shaking fall off with distance from the source?
The hypothesis. Written **beforehand**: from five metres the amplitude will be _____ times smaller than from one metre. Not changed afterwards.
What I change. One thing only: the **distance** the strike is made from.
What I measure. The peak amplitude, in g, from the card, **and the noise floor** measured

before the series.
What I keep the same. The same strike — the same object, the same drop height or the same fist, the same person — the same table, the same box position, the same silence in the room. **The hardest thing to keep the same is how hard you strike**, which is why a book dropped from a marked height is far better than a fist.
How many repeats. Three strikes at each distance, with silence between them, and take the mean. A single strike says nothing: the spread between three "identical" strikes is itself a result, and usually a surprisingly large one.
What makes a reading valid — and what makes it an artefact. Valid is a reading taken after ten seconds of silence, with nobody touching the table, and **only if the peak clearly exceeds the noise floor**. An artefact is anything else: an elbow on the table, a hanging cable pulling the box, a classmate walking past, **or a strike so hard that the peak runs off the ±0.25 g scale**. The class rule: announce, count ten, strike, read.

6. Analysis after the activity

6.1 What the data should look like

Source	Typical amplitude	What it means
Full silence, good table	about 0.01 g	**The noise floor**, mostly the chip's own electronic noise. The threshold below which nothing is detected.
Silence, but the class standing	0.02–0.05 g	A noise floor two to five times higher; comparisons become impossible.
Light finger tap, 1 m	0.02–0.06 g	Above the floor, but not by much.
Fist, 1 m	0.05–0.20 g	Close to the edge of the fixed scale.
The same knock, 5 m	0.01–0.03 g	Several times smaller, often right at the edge of the noise. **The central result of the activity.**
Normal footstep, 1 m	0.02–0.08 g	Surprisingly large.
Door closed normally, 5 m	0.01–0.03 g	At the limit of detection — which is exactly why nobody leaves the room during a measurement.
Box on foam, same knock	30–70% smaller	Attenuation by a soft foundation.

6.2 Questions to ask afterwards

1. The same knock gave two different numbers, from one metre and from five. Which describes *the knock*, and which describes *what was felt at the box*?
2. The shaking index showed 2 for a fist on the table. Why does that not mean a magnitude-2 earthquake? What is missing from the calculation?
3. Why are real seismographs installed in basements, on bedrock and far from roads? Use your noise-floor measurement as the argument.
4. Three "identical" strikes gave three different numbers. What does that tell you about how reliable a single measurement is?
5. The box on foam read less than the box on the table for the same strike. What do engineers

do with that idea when designing buildings in seismic zones?

6.3 Assessment ideas

- **Low stakes:** the pupil records the noise floor and ranks the five sources by amplitude.
- **Standard:** the pupil compares every source with the noise floor, computes how many times the amplitude falls between one metre and five, and writes a sentence about why.
- **Extension:** the pupil explains in writing the difference between magnitude and intensity, using their own data as the example, and argues why the app's shaking index actually measures the second.

Common pitfalls and misconceptions

Skipping the calibration. Without a measured noise floor, no peak can be declared "detected". It is the first measuring step and it is not skipped.
Expecting a floor of "zero". The sensor alone has a noise of about ±0.01 g, however quiet the room. It is not a fault and it cannot be lowered; it is measured and declared.
The class standing. Thirty people on a springy floor raise the noise floor tenfold. **Everybody sits down during a measurement**, including those not watching.
Striking the box, or the table right under it. You strike the table at the marked distance, never the box. A direct blow saturates the sensor and measures nothing useful.
The hanging cable. A USB cable hanging off the table edge pulls the box and adds vibrations of its own at every movement of the laptop. Tape it down or work over WiFi.
Taking the shaking index for a magnitude. It is a relative scale with no distance correction. Do not compare it with real magnitudes and do not let pupils write "we detected a magnitude 2 earthquake".
Strikes that run off the scale. The scale is fixed at ±0.25 g. A very hard knock looks identical to an even harder one, because both reach the edge. Strike more gently.
One strike per distance. The spread between "identical" strikes is large. Three strikes and a mean.
Measuring immediately after starting. The filter needs a few seconds of quiet to learn the zero line. Begin with the silence, not with the strike.

Safety and classroom management

The box is not struck. The table is struck, at the marked distance. The box is school equipment, not a target, and a direct blow can damage it.
Nothing heavy is used to strike. A fist, a pencil, a book dropped from 20 cm — that is all. Nothing metal, nothing thrown, and nobody kicks the table.
Doors. If you use a closing door as a source, **the teacher** closes it, normally, once. A slammed door catches fingers and adds nothing to the experiment.
Moving about the room. The activity requires pupils to walk to marked distances. Agree the route and the rule — walk, do not run — before anybody stands up.
Cables. A cable stretched from the laptop to the measuring table is a trip hazard and spoils the measurement besides. WiFi solves both.
The conversation about earthquakes. In Romania some pupils have lived through a strong earthquake, or have relatives who did. **Announce the topic in advance**, treat it as a matter of preparedness rather than fear, and leave room for questions; the *Earthquake!* activity continues exactly this discussion, with the safe-behaviour part.

Adaptations for pupils with special educational needs

Ranking instead of numbers. The main adaptation: the pupil records no values in g but **ranks the five sources** from weakest to strongest, using the shape of the wave on screen. The objective — that shaking falls off with distance and depends on the source — is met in full with no figures.
The graph read, not measured. For many pupils the *shape* of the seismogram — a sharp peak dying away — says more than a number. The sheet has a box for drawing the shape of each source.
The role of "silence keeper". This is the most valuable role in the class and it is done from a desk: the pupil announces the start of the measurement, counts the ten seconds aloud and **has the right to void a measurement** if anybody touched the table. **For a pupil who needs a clear task and real authority, this is the role**, and they genuinely decide which data are valid.
Two more real roles. The screen reader says the peak amplitude aloud; **the scribe** writes it down and computes the mean of the three strikes.
Noise and surprise. The strikes are sudden and loud. Announce each one before making it — "striking now" — which the protocol requires anyway, and seat pupils for whom sudden noises are unpleasant further from the table.

If things do not go to plan

The table wobbles at every footstep. Move the box to **the floor**, directly on concrete or tiles, in a corner of the room. A good floor beats any bad table. If the floor is wood on joists, try beside a load-bearing wall, where it moves least.
You cannot get silence. Measure the noise floor *as it is* and declare it. The activity remains valid, but you will only be able to detect the large sources — which is exactly the situation of a seismograph installed in a city, and worth saying.
Every strike runs off the scale. Strike more gently or move further away. The fixed ±0.25 g scale is designed for small vibrations, not for blows.
Nothing is detected, not even a fist. Check that you are striking the table the box is standing on — a blow on another table, even an adjacent one, transmits far more weakly. Check too that the activity is receiving data: the line must visibly tremble, even in silence.
The shaking index shows zero. Below the calm threshold (about 0.012 g) the app displays 0 rather than a number. It is not a fault: it means the peak is too close to the noise floor.
The box will not connect. Do not stop the lesson. Use the "What the data should look like" table as a printed dataset: pupils rank the sources, compute how many times the amplitude falls from one metre to five, and write the conclusion about magnitude and intensity. **The whole conceptual part — which is the lesson — is done without the equipment.** Measure again another day, after checking the cable, the port selected in the app and, if you work over WiFi, the network.

7. Further resources

Stable, well-established starting points. Search within each site for the topic named — these are base addresses, not deep links, so they do not rot.

- **Romanian National Institute for Earth Physics** — `https://www.infp.ro` — the Romanian earthquake catalogue and the intensity maps. **The most relevant source for a Romanian class**, and where pupils can see the Vrancea earthquakes of the last few days.
- **USGS Earthquake Hazards** — `https://www.usgs.gov` — search for magnitude and intensity; the classic explanation of the difference, with "did you feel it" maps that are intensity

(the Modified Mercalli scale used in the USA) collected from people.

- **IRIS / EarthScope** — `https://www.iris.edu` — teaching material on seismograms and how to read them, plus real seismograms to download.
- **Institute of Physics** — `https://www.iop.org` — search for material on waves and inertia, for the physics half.
- **PhET Interactive Simulations** — `https://phet.colorado.edu` — search for the wave simulations; useful for showing the geometrical spreading pupils have just measured.
- **Romanian General Inspectorate for Emergency Situations** — `https://www.igsu.ro` — for the preparedness and behaviour part, which continues in the *Earthquake!* activity.

8. Assessment

The exit ticket, identical to the one on the pupil sheet: *the same knock gave two different numbers. Why are both correct?* Collected in the last two minutes, it says more about the class's understanding than any mark.

For a more formal judgement, use the **common three-axis rubric** — *measurement practice*, *data interpretation* and *scientific reasoning*. At Explorer level the three axes look like this:

- **Measurement practice** — the pupil measures the noise floor before anything else, respects the silence, repeats every strike three times, and does not strike the box.
- **Data interpretation** — the pupil compares every peak with the noise floor, computes the ratio between distances, and recognises a saturated measurement.
- **Scientific reasoning** — the pupil distinguishes magnitude from intensity, explains why the shaking index measures the second, and links the fall-off with distance to why seismographs stand on bedrock.

The rubric itself, with descriptors for all three levels, is defined once in the *curriculum plan* and is not repeated in the guides.

9. Pupil sheet

The sheet the pupils fill in is a separate document: `docs/en/student_sheets/seismometer.pdf`. It has two pages — the noise floor, the prediction and the table of sources on the first; the distance series, the drawing of the wave shape and the conclusion on the second — so it is printed double-sided, one sheet per pupil. It is built on the three layers — what I see on the screen, what I do on my own, the conclusion I write.

Three things to know before printing it. **The noise floor has a large box of its own at the head of the page**, and every comparison refers to it: a sheet with the floor left blank cannot support any claim of the form "we detected". **The table asks for three strikes and a mean**, printed as such, because the spread between "identical" strikes is large and a single value misleads. **The box for drawing the wave shape sits beside the table**, because for many pupils the shape — a sharp peak dying away — says more than the figure, and on the support route it replaces the number column entirely. The tables are not repeated here precisely so that the two documents cannot drift apart: there is one printable version, and it is the sheet.